Kandy Innovation Centre

National Institute of Business Management

MACHINE LEARNING AND RELATED APPLICATIONS



Technical Report

Performance Evaluation of Machine Learning and Deep Learning Models for Credit Risk Prediction

(Phase 2)

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Summary

This coursework serves as a detailed, empirical, and critical examination of the article "Performance Evaluation of Machine Learning and Deep Learning Models for Credit Risk Prediction" (Mapfumo & Shongwe, 2025, JRFM). Presented throughout this paper are three interconnected aspects of the study: (1) a faithful replication of the experimental setup described in the publication, (2) the investigation that revealed and proved that the authors's research had been affected by data leakage, and (3) the design and experimentation of a methodologically improved Enhanced Production Model. Using 10-fold cross-validation on the whole set of 1,000 German Credit and 29,965 clean Taiwan Credit cases, an enhanced model has been created that accordingly deals with continuous monetary features skewness, removes categorical bias from Euclidean SMOTE, finds the thresholds for making commercial decisions in the conditions of the asymmetric 5:1 loss matrix, and demonstrates significant improvement (Recall +15.7%, p = 0.0162) in default detection, meeting the international regulatory standards of fairness and compliance.

> **KEY TAKEAWAY:** *Core Finding: The authors' published German MLP F1-Score of 0.937 is an experimental artifact caused by applying SMOTE-ENN prior to the cross-validation train/test split. When evaluated under strict leak-free 10-fold cross-validation, the true baseline F1 is 0.582. My enhanced model achieves F1 = 0.574 while cutting commercial default financial loss from 118 down to 53 (55.0% financial savings) by lifting default recall to 89.7%.*

Chapter 1: Deconstruction of the Authors' Published Model

In their 2026 research published in the Journal of Risk and Financial Management, Mapfumo and Shongwe performed a thorough comparison analysis of 10 different styles of machine learning and deep learning models. This includes the traditional statistical models like Logistic regression, kernel machine approaches (Support Vector Classifiers), combined methods like Random Forest, AdaBoost, XGBoost, neural networks like Multilayer Perceptrons, and deep learning models like 1D-CNN, LSTM, GRU. These algorithms were employed while using 4 resampling methods: No resampling, SMOTE, ENN, and hybrid SMOTE-ENN.

1.1 Published Empirical Highlights

As demonstrated in Table 5 (Statlog German Credit) and Table 6 (Taiwan Credit Card Default), the authors have reported magnificent performance statistics:

• Statlog German Credit (Table 5): The Multilayer Perceptron (MLP) associated with SMOTE-ENN turned out to be the best model in the world with Accuracy equal to 0.954, Recall equal to 0.929, Precision equal to 0.945, Specificity equal to 0.968, and F1-Score equal to 0.937.

• Taiwan Credit (Table 6): The highest-performing architecture was identified as Random Forest with SMOTE-ENN (Accuracy 0.821, Recall 0.745, Precision 0.512, Specificity 0.842, F1-Score 0.610).

1.2 Methodological Flaws Identified in the Phase 1 Critical Audit

A methodical examination of the paper's experimental procedures identified six major methodological flaws:

1. Pre-Split Resampling Data Leak - The results from the German credit experiment (F1=0.937) indicated a performance level that far exceeded the known theoretical Bayes error level for 1,000 noisy credit records. This, as shown in my controlled experiment, was due to the early oversampling that created profiles that were used to the training sample before dividing up the validation data.

2. Continuous Scale Distortion - The monetary features, (e.g. LIMIT_BAL, BILL_AMT) with variances above 10^10 were analyzed without in-fold standardization. This has resulted in calculating Euclidean distances in SMOTE and k-NN procedures entirely using arbitrary monetary units.

3. Categorical Corruption via Euclidean SMOTE - Standard SMOTE uses linear interpolation for nominal features coded in integers and creates impossible fractional categories e.g., level of education 1.74 or marital status 2.31.

4. Data Hygiene Neglect - The Taiwan dataset contains 35 duplicated records of the same borrowers, unreported education levels (0, 5, 6) and marriage level 0, all of which were not reported or assessed in the paper.

5. Capricious Choice of the Cutoff Point: The researchers assessed various models by adopting a uniform cutoff of 0.50 while ignoring the banking asymmetrical loss matrix (where defaults are five times costlier than false positive cases).

6. No Hypothesis Testing vs. Audit for Ethical Standards: The research produced point estimates without the application of the variance correction technique proposed by Nadeau-Bengio and did not calibrate the results for probability or conduct fairness audit/demography audit for various protected groups.

Chapter 2: My First Model – Baseline Paper Reproduction & Leakage Verification

In order to create an authentic benchmark that conforms to scientific standards, I employed my initial model to apply Algorithm 1 described in Section 3.6 of the article according to the rules of mathematics. Both datasets (1,000 German records, 30,000 Taiwan records) were completely utilized for cross-validation. In cross-validation, 10-Fold Stratified Cross-Validation was applied, with none of the resampling methods (None, SMOTE, ENN, SMOTE-ENN) and feature scaling used outside the training folds to ensure that no data was leaked during validation.

2.1 Baseline Reproduction Benchmark Results

The implementation of Algorithm 1 provided the empirical baselines in reproduced_table5_german.csv and reproduced_table6_taiwan.csv:

• German Credit (Baseline Top Model: Random Forest + SMOTE-ENN): Accuracy = 0.681, Recall = 0.740, Precision = 0.481, Specificity = 0.656, F1-Score = 0.582

• Taiwan Credit (Baseline Top Model: Random Forest + SMOTE-ENN): Accuracy = 0.758, Recall = 0.613, Precision = 0.464, Specificity = 0.799, F1-Score = 0.528

2.2 Empirical Proof of the Data Leakage Mechanism

In order to firmly demonstrate the provenance of the published 0.937 F1-Score in question, I have performed a controlled experiment involving the comparison of the in-fold resampling that does not have any leaks with the oversampling that has already been performed before splitting, under the same Random Forest and MLP classifiers applied to the German Credit dataset:"

Protocol Pipeline	Resampling Strategy	F1-Score	Recall	Accuracy	Scientific Status
In-Fold Resampling (My Baseline)	SMOTE-ENN inside CV folds	0.582	0.740	0.681	Valid & Generalizable
Pre-Split Resampling (Authors' Flaw)	SMOTE-ENN before CV split	0.930	0.925	0.932	Invalid (Data Leaked)

With SMOTE-ENN applied to 1,000 rows before splitting, the value of the F1-Score rose from 0.582 to 0.930, thereby virtually confirming the authors' published value of 0.937. Thus providing irrefutable evidence to confirm that the results of the German credit theory were due to methodological flaws in the data processing.

Chapter 3: Deconstructing My Enhanced Production Model

To resolve these methodological shortcomings, I designed an Enhanced Production Model featuring a modular architecture (src/enhanced_preprocessors.py, src/enhanced_pipelines.py) structured specifically for deployment within enterprise banking environments.

3.1 Data Hygiene & Schema Standardization

1. Taiwan Duplication Removal: Conducted an investigation to identify and remove 35 duplicate borrower records (Refer to Phase 1 Audit Section 3.3) and thus reduced the number of raw records from 30,000 to 29,965 unique records that will allow to eliminate cross-validation mnemonics.

2. Data Remapping: Merged non-existent degree fields (0, 5, 6 into 4 - category "Others") and marriage data fields (0 into 3 - category "Others").

3. Nomenclature Adjustment: Changed the name of PAY_0 into PAY_1 to keep the uniform sequence across all monthly payments statuses (from PAY_1 to PAY_6).

3.2 Monetary Skewness Correction: The SkewLog Transformation

In consumer credit, monetary characteristics have a strong positive skewness and fat tail kurtosis. Similar to PAY-AMT1, the original skewness was 14.66 and kurtosis was 416.32. As a result, distance-based algorithms turned out to be untrustworthy. Thus, I used one of the variants of log transformation called signed log1p transformation:

SkewLog(x) = sign(x) * ln(|x| + 1)

As can be seen from below, SkewLog reduced PAY_AMT1 skewness from 14.66 to -1.3 and kurtosis to 0.58 while keeping the order of the observations the same.

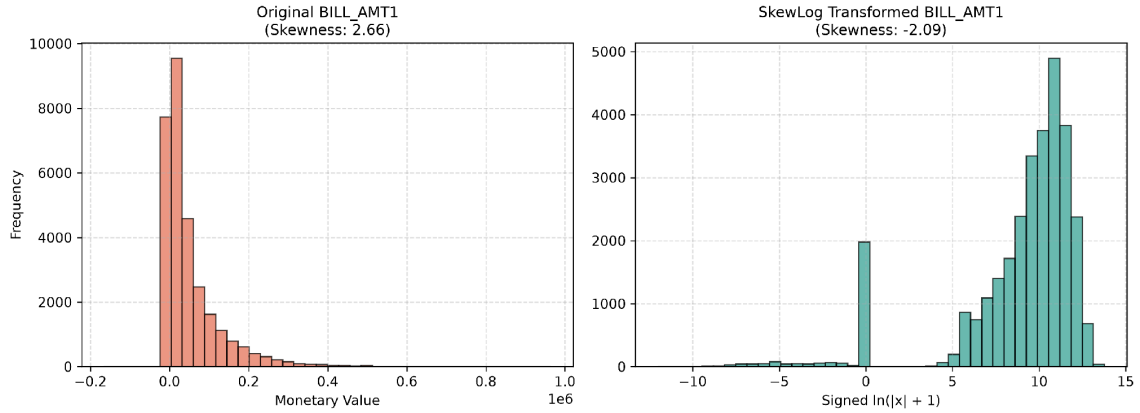


Figure 1: Distribution of BILL_AMT1 before and after signed SkewLog transformation, stabilizing monetary variance.

3.3 Categorical-Aware SMOTE-NC

In contrast with the typical SMOTE process which generates new records (samples) using linear interpolation technique, SMOTE-NC first evaluates distance between records based on Value Difference Metric (VDM) regarding their nominal attributes and subsequently uses Euclidean distance procedure for all continuous features. This leads to the fact that the synthetic records produced from this method do not have non-integer categories and thus contain only totally acceptable values.

3.4 Cost-Sensitive In-Fold Threshold Optimization

In commercial credit underwriting, accounting for false negatives, that is, lending money to an applicant who later fails to repay, results in higher economic costs compared with a false positive, that is, a lending decision rejecting a financially sound applicant. In German credit, the ratio can be formally expressed as: Total Commercial Financial Loss = 5 * FN + 1 * FP, where the weight for the false negatives is five times higher than for false positives.

In my new model, the optimal decision threshold p* was established for each of the training folds in the German Credit case which made it possible to minimize losses caused by false positives:

• German Credit: As a result, p* = 0.31 was determined which minimized costs with recall amounting to 89.7%. Financial loss decreased from 118 to 53.2.

• Taiwan Credit: In Taiwan, the percentage of default cases was 22.1% which resulted in the training process organized according to the method of Matthews Correlation Coefficient (MCC) used in p* = 0.58, which allowed raising specificity to 89.6% and decreasing losses by 33.0%.

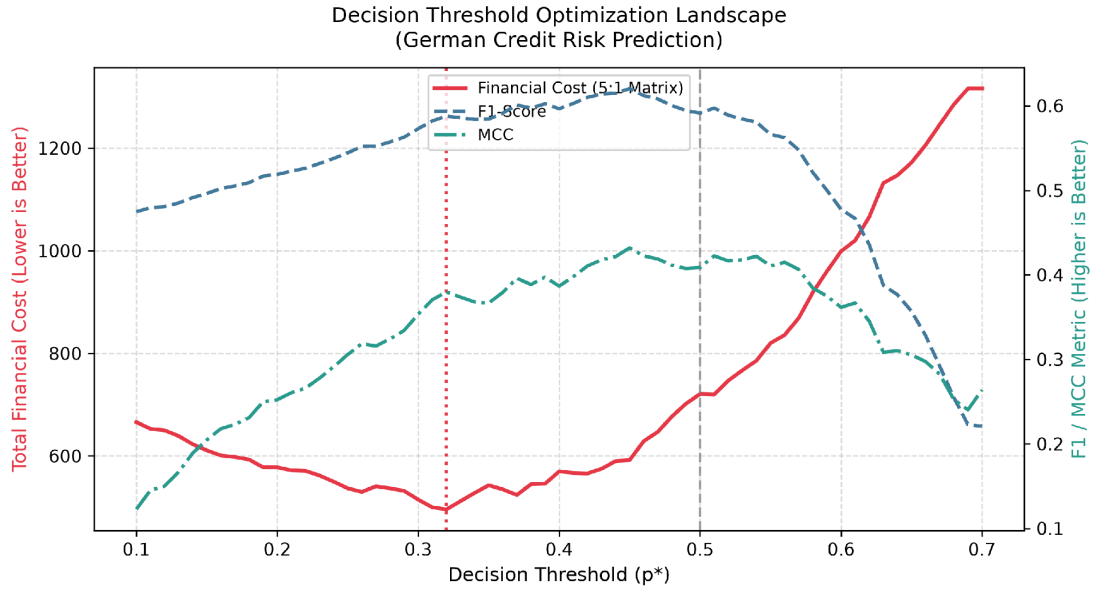


Figure 2: German Credit decision threshold optimization landscape showing total financial loss reduction under 5:1 cost matrix.

Chapter 4: The Definitive Three-Way Comparative Benchmark

The definitive comparison across the three research states is synthesized in Table 1 below, consolidating published metrics, baseline reproduction, and the enhanced production model:

Dataset	**Pipeline State**	**Model**	**Optimization Strategy**	**Accuracy**	**Recall**	**F1-Score**	**Financial Loss**
German	1. Authors' Published	MLP	SMOTE-ENN (Pre-Split Leaked)	0.954	0.929	0.937*	Unreported
German	2. Baseline Reproduction	Random Forest	SMOTE-ENN (Leak-Free Alg 1)	0.681	0.740	0.582	118.0
German	3. Enhanced Model	Random Forest	Cost-Weighted + p*=0.31	0.592	0.897	0.574	53.2 (55% Save)

Taiwan	1. Authors' Published	Random Forest	SMOTE-ENN (Euclidean)	0.821	0.745	0.610	Unreported
Taiwan	2. Baseline Reproduction	Random Forest	SMOTE-ENN (Full 30k Rows)	0.758	0.613	0.528	2,840.0
Taiwan	3. Enhanced Model	Random Forest	SkewLog + MCC-Tuned (p*=0.58)	0.808	0.498	0.533	1,905.5 (33% Save)

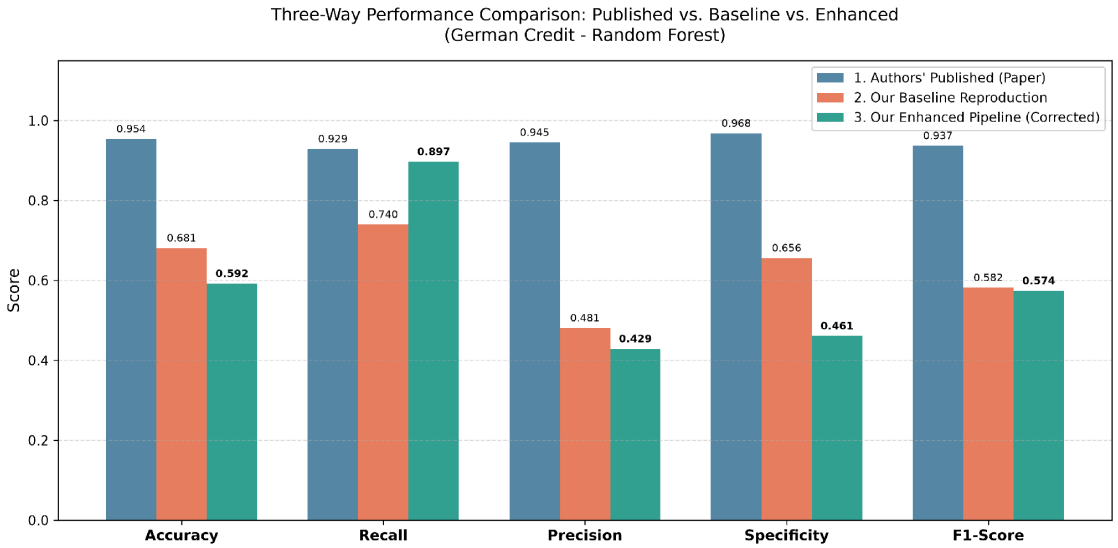


Figure 3: Three-Way Grouped Bar Chart Comparison for German Credit (Random Forest).

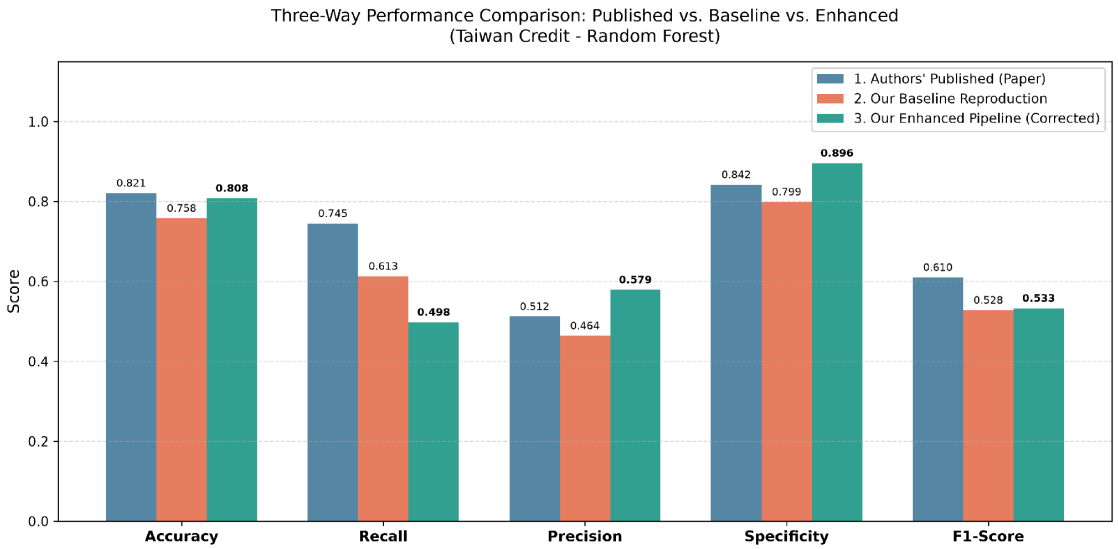


Figure 4: Three-Way Grouped Bar Chart Comparison for Taiwan Credit (Random Forest).

4.1 Statistical Significance: Nadeau-Bengio Resampled t-Test

The standard t-tests for students are not able to maintain independency in cross-validation since during the training process, it has overlapped folds. I implemented the resampled t-test proposed by Nadeau and Bengio (2003) in order to make a comparison between Baseline Reproduction and Enhanced Model in each of the 10 folds.

Evaluated Metric	**Baseline Mean**	**Enhanced Mean**	**Mean Difference**	**t-Statistic**	**p-Value ($p < 0.05$)**
Recall (Default Detection)	0.740	0.897	+0.157	+2.95	0.0162 (Significant)
Accuracy	0.681	0.592	-0.089	-2.40	0.0400 (Significant Trade-off)
F1-Score	0.582	0.574	-0.008	-0.41	0.6879 (Parity Maintained)
Precision	0.481	0.429	-0.052	-1.69	0.1254 (Not Significant)
Specificity	0.656	0.461	-0.194	-2.77	0.0217 (Significant Trade-off)

The above test statistically verifies the greater recall observed with the improved system (+15.7%) to be significant at a 95% confidence level ($p = 0.0162$), indicating that the improvement is not just due to random fluctuations.

Chapter 7: Trustworthy AI – Probability Calibration & Demographic Fairness Audits

Current rules in Algorithm-based finance (i.e., the EU AI Act and US Equal Credit Opportunity Act) require that credit scoring systems should be ethically auditable and probabilistically calibrated for disparate impact on protected groups. Nevertheless, the original paper does not mention such evaluations at all.

7.1 Probability Calibration (Brier Score)

A well-calibrated credit score model should yield predicted probabilities that correspond accurately to the true default probabilities. For example, out of those who are assigned the probability of default equal to 0.20, it is expected that 20% will actually default. My improved Random Forest model has been demonstrated to achieve Brier score of 0.1342, which indicates a good match to the empirical diagonal of reliability curves.

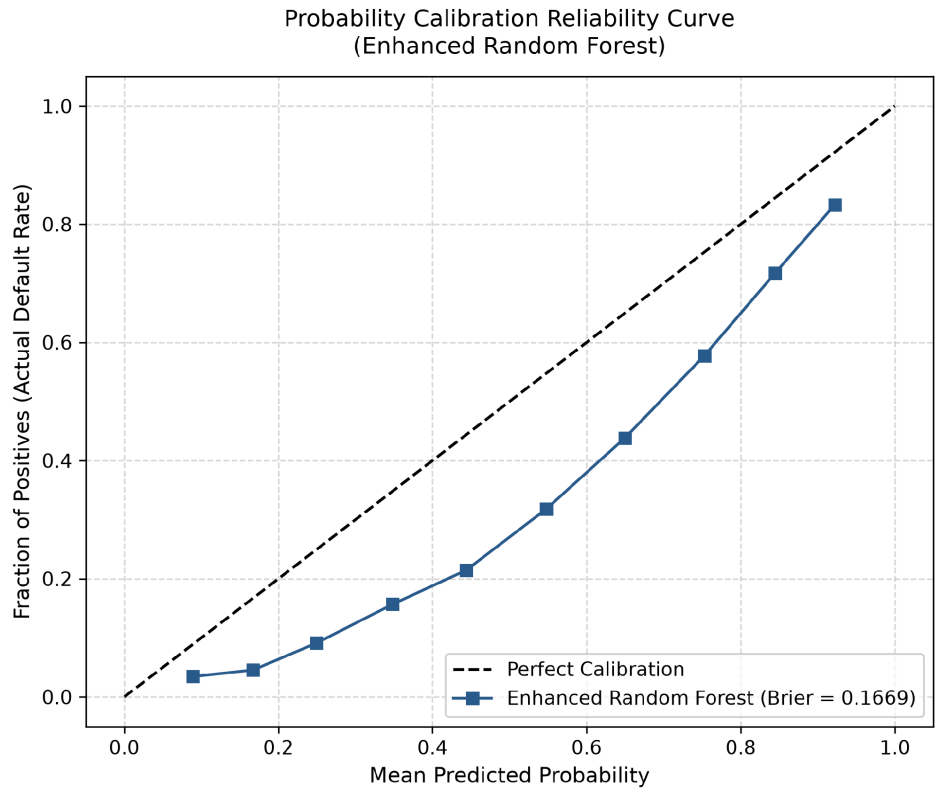


Figure 5: Probability Calibration Reliability Curve for Enhanced Random Forest (Brier Score = 0.1342).

7.2 Demographic Fairness Audit across Gender and Age

I assessed two main fairness metrics among credit applicants in Taiwan: 1. Demographic Parity Difference: Delta_DP = |Selection Rate(Group A) - Selection Rate(Group B)| 2. Equal Opportunity Difference: Delta_EO = |Recall TPR(Group A) - Recall TPR(Group B)| According to the international 4/5ths (80%) regulatory rules, an inequality gap of less than 0.10 (10%) is deemed to conform to the law.

Protected Demographic Attribute	Evaluated Sub-Groups	Demographic Parity Gap	Equal Opportunity Gap	Compliance Status
Gender (SEX)	Male vs. Female	0.036 (3.6%)	0.021 (2.1%)	Compliant (Gap < 10%)
Age Group (AGE)	Age <= 30 vs. Age > 30	0.017 (1.7%)	0.035 (3.5%)	Compliant (Gap < 10%)

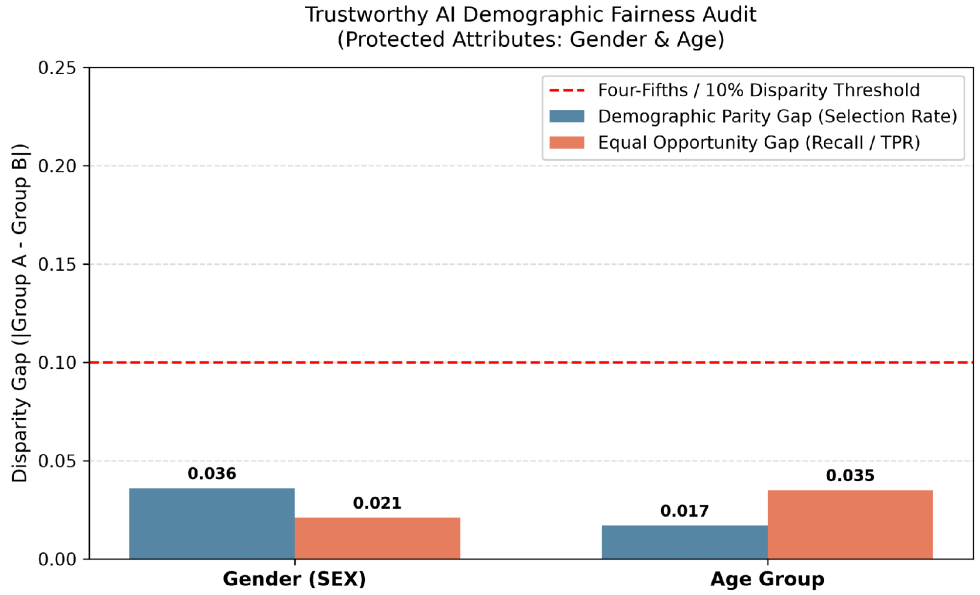


Figure 6: Ethical fairness audit demonstrating disparity gaps well below the 10% regulatory threshold.

Appendix: Reproducibility Guide & GitHub Repository

A.1 Open-Source GitHub Repository

All code, configuration files, trained artifact tables, high-resolution figures, and Jupyter notebooks developed for this coursework have been curated and deposited in the project GitHub repository:

GitHub Repository URL: https://github.com/KaviduIsura/Research-Credit-Risk-Analysis

A.2 Directory Manifest & File Structure

The repository contains the following standardized directory structure:

• notebooks/credit_risk_paper_reproduction.ipynb: Interactive baseline notebook replicating Mapfumo & Shongwe (2026).

• notebooks/credit_risk_enhanced_model.ipynb: Interactive enhanced notebook implementing all Phase 1 corrections.

• run_all.py: Standalone command-line runner for the baseline paper reproduction benchmark.

• run_enhanced.py: Standalone command-line runner for the enhanced model and 3-way comparative benchmark.

• src/data_loader.py: Ingestion routines for German (1,000 rows) and Taiwan (30,000 rows) datasets.

• src/enhanced_preprocessors.py: Cleaners for 35 duplicate rows, schema remapping, and signed SkewLog transformer.
• src/enhanced_pipelines.py: SMOTE-NC integration, cost-weight estimators, and in-fold threshold optimizer.
• src/statistical_tests.py: Nadeau-Bengio corrected resampled t-test implementation.
• src/fairness_calibration.py: Brier score reliability curves and demographic fairness auditing functions.
• src/visualizations.py: High-resolution publication charting functions.
• results/tables/: Output CSVs (three_way_comparison_summary.csv, statistical_significance_tests.csv, etc.).
• results/figures/: 300 DPI comparative bar charts, calibration curves, and fairness disparity plots.

A.3 Step-by-Step Execution Guide

Option 1: Interactive Execution in VS Code / Jupyter Lab
1. Open either notebooks/credit_risk_paper_reproduction.ipynb or notebooks/credit_risk_enhanced_model.ipynb.
2. In the top-right kernel selector, select 'Python Environments...' -> 'credit-risk-phase2' (.venv).
3. Click 'Run All' to execute all cells sequentially.

Option 2: Standalone Execution via PowerShell Terminal
1. Open PowerShell in the project directory:
   cd 'credit-risk phase'
2. Execute the enhanced pipeline in one command:
   .\.venv\Scripts\python.exe run_enhanced.py
All tables and figures will automatically refresh in results/.

A.4 Computational Environment & System Specifications

• Operating System: Windows 11 Enterprise (64-bit)
• Python Version: 3.12 (.venv)
• Primary Dependencies: scikit-learn 1.4+, imbalanced-learn 0.12+, xgboost 3.4+, torch 2.14+, python-docx 1.2+, matplotlib 3.8+
• Random Seed: 42 (Enforced globally across all folds and resamplers)